

Privacy-aware Human Activity Recognition with Smart Glasses for Digital Therapeutics

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ABSTRACT

Human Activity Recognition (HAR) holds valuable potential as a digital biomarker, offering insights into individuals' physical activities and well-being. Leveraging wearable devices to monitor human activities can serve as an integral component of digital therapeutics, offering valuable insights into an individual's health and behavior. However, privacy concerns in HAR systems pose a significant challenge as users' data can be at risk of exposure and necessitates robust protection measures. To mitigate these concerns, the adoption of privacy-preserving techniques, such as federated learning, becomes necessary. This paper addresses the topic of Privacy-aware Human Activity Recognition (HAR) with Smart Glasses and explores a federated learning approach to achieve accurate recognition while preserving data privacy. The study highlights the trade-off between model complexity, recognition performance, and data privacy. Initially, a comparison was conducted between a complex deep learning network (STResNet) and a simpler deep learning network (DANA). The complex network exhibited an F1-Macro score of 0.87, while the simpler network achieved 0.82. Subsequently, the simpler network was further evaluated by comparing it to its federated learning implementation. The results revealed that the simpler network attained an F1-Macro score of 0.85, whereas federated learning achieved a slightly lower score of 0.80. These findings shed light on the performance disparities between the different approaches.

CCS CONCEPTS

• **Applied computing** → Life and medical sciences; Health informatics.

KEYWORDS

Human activity recognition, Privacy, Federated learning, Smart glasses, Digital biomarker

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1 INTRODUCTION

Human Activity Recognition (HAR) refers to the process of automatically identifying, classifying, and understanding human activities based on data captured from various sensors, such as accelerometers and gyroscopes. It has gained significant attention as a digital biomarker with promising applications in mental health and various healthcare domains [1]. Monitoring human activities with wearables can be used as a component of digital therapeutics. The Digital Therapeutics Alliance (DTA) defines digital therapeutics as “evidence-based therapeutic interventions that are driven by high-quality software programs to treat, manage, or prevent a disease or disorder” [2]. Data from wearable devices can provide valuable insights into an individual's health and behavior, which can be used to tailor digital therapeutic interventions. For example, a digital therapeutic app may utilize activity data to create personalized exercise plans or provide reminders for medication adherence. By integrating wearable monitoring, healthcare providers can gain objective information for precise and individualized treatments. HAR has the potential to assist in early detection, monitoring, and management of various conditions, including neurodegenerative

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disorders like Parkinson’s disease [3], Schizophrenia [4], mental health disorders [5], and elderly care [6]. However, privacy concerns pose significant challenges when dealing with personal data collected through wearable devices [7]. Wearable devices capture sensitive information, including the movements, behaviors, and routines of individuals. Traditional data collection approaches often involve streaming data to a central server or cloud, where it is processed and analyzed. This centralized approach raises privacy concerns and increases the risk of data breaches or unauthorized access, potentially compromising individuals’ confidentiality and undermining their trust in using such technologies. To address the privacy issue, we propose a privacy-aware system that leverages federated learning, a distributed learning framework that ensures data privacy while enabling model training and improvement. Federated learning allows models to be trained directly on user devices without requiring raw data to leave the device, thus preserving the privacy of the data contributors. Prior studies have explored various techniques for HAR and privacy preservation [8]. Some approaches involve data anonymization by utilizing homomorphic encryption (HE) [9, 10]. In computer vision HAR, the images and videos are augmented in a way that will preserve the privacy of the users by processing the videos in special low-quality conditions [11], cartoon like effects [12] etc. Sozinov et. al combine state-of-the-art deep learning models with federated learning for predicting human activities based on accelerometer and gyroscope data [13]. Sannara Ek et al. compare multiple federated learning algorithms on different datasets that contain data from smartphones [14]. Shen et al. propose a novel federated meta-learning approach for diversity aware HAR [15].

In this paper, we compare three different approaches for HAR: a complex deep learning (DL) architecture (STResNet) [16, 17], a simpler DL architecture (DANA) [18], and the implementation of the simpler DL architecture with federated learning. These approaches aim to recognize and classify activities based on data collected with emteq’s OCOsense™ [19, 20] multi-modal smart glasses. By examining the performance of these approaches, we aim to assess the effectiveness of federated learning as a privacy-preserving method for HAR and explore its benefits and limitations compared to the other approaches.

2 DATASET

The data collection process 18 young adult research volunteers, evenly split between 9 males and 9 females. The age range of the participants spanned from 18 to 37 years, with an average age of 23 ± 5 . These individuals were selected without strict criteria for participation. Eligibility was determined by the ability to comfortably wear the one-size-fits-all glasses, ensuring a diverse sample size not limited by factors such as age, gender, or physical ability. Before the start of the data collection all participants provided written informed consent. To capture the data, each participant was equipped with OCOsense™ smart glasses (Figure 1). These glasses feature various sensors, including a 9-axis inertial measurement unit (IMU) sensor comprising an accelerometer, gyroscope, and magnetometer, as well as a pressure sensor. For this study, only the accelerometer and gyroscope data were utilized, as they provided sufficient information about the participants’ movements while maintaining the

size of the data. The sensors continuously collected data at a fixed rate of 50Hz.

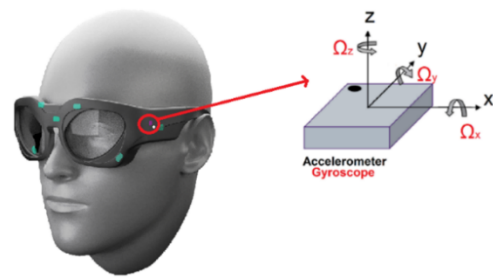


Figure 1: OCOsense™ smart glasses. The green colored rectangles represent the OCO™ sensors. The 9-axis IMU is represented by the purple rectangle.

Real-time data streaming from the smart glasses to an iPad application was enabled through Bluetooth. Simultaneously, a researcher conducted real-time data labeling. To enhance the precision of activity labeling, the dataset also included synchronized video data from five external cameras positioned strategically throughout the apartment.

Prior to the data collection procedure, participants received instructions on how to properly wear the glasses to ensure a comfortable fit. Adjustments were made as needed to guarantee optimal comfort during the data collection process. The procedure involved three predefined scenarios that encompassed a wide range of activities of interest. Participants were provided instructions regarding these scenarios but were given the freedom to perform the activities in their own preferred manner. As a result, the activities were categorized into three groups: posture activities (e.g., sitting, standing, lying, walking, and transitions between sitting and standing), personal hygiene activities (e.g., washing hands and brushing teeth) and dietary intake activities (e.g., eating and drinking). Each participant’s activities were recorded for a duration of 2 hours, leading to a comprehensive dataset with a total of 36 hours of labeled data. For this study, the analysis focused on four key activities, including sitting, standing, lying and walking, because we aim to detect only the posture related activities and reduce model complexity. The distribution of the data across these activities can be observed in Figure 2.

3 METHOD

Prior to inputting the data into the chosen deep learning architectures, we applied a fifth-order median filter to the signals for signal denoising. Following the filtering process, we computed the magnitude values for both the accelerometer and gyroscope sensors. Determining the appropriate window size for segmenting the data was the subsequent step in the procedure. To identify the optimal window size, we conducted experiments with various window sizes. Through empirical analysis, we determined that a window size of 4 seconds provided enough temporal information for the models to capture the patterns between different activities. Consequently, the signals were segmented using a window size of 4 seconds, with a 1-second slide between consecutive windows. After

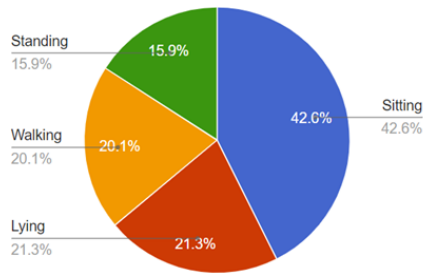


Figure 2: Activity distribution in the dataset

this simple preprocessing step, we first utilized a complex deep learning network, STResNet (Spectro-Temporal Residual Neural Network). This network extracts temporal and frequency information for each sensor through channel specific residual blocks and spectrograms. Although this architecture performs well, it has many parameters (40 million) which makes it difficult for on-device implementation. For this reason, a simpler deep learning architecture – Dimensionality Adaptive Neural Network (DANA) consisting of 2D convolutional layers (CNNs), LSTM layers and a fully connected layer was proposed. Then we explored the performance of the two architectures and the drawbacks from using a simpler architecture. Cross entropy was used as the objective function for training with Adam as an optimizer with dynamic learning rate starting at $1e-3$ and decreasing by a factor of 0.1 every fifty epochs. The total number of epochs for both architectures was set to 200. To prevent overfitting, we employed an early stopping callback that monitors the F1-macro score. F1-macro score was monitored because it provides unweighted mean of the F1-Scores for each of the classes.

The OCOsense™ glasses stream data to an iPad application which stores the data in cloud (Figure 3).

To address the concern of exposing sensitive data during the training of models on cloud data, our third approach introduces an innovative solution to ensure secure and private HAR by leveraging federated learning. This approach allows us to train the simpler DL model mentioned earlier while preserving the privacy of the data. By adopting federated learning, the model is trained on decentralized data without the need for data to be transmitted to a central server, thereby ensuring the confidentiality of individual user data. We used Flower library to implement federated learning. Each of the 18 participants in the dataset was treated as a client. The clients participated in the training process which lasted 10 rounds. During each round, 70% of each client's data are used for model training, while 30% of each client's data is used for evaluation purposes, ensuring a comprehensive assessment of the model's performance. Furthermore, a comparative analysis was conducted between the performance of the federated learning model and the simpler DL network to assess the potential drawbacks associated with adopting a privacy-aware approach. To preserve computation time, we used 20 epochs per round in the FL approach and also 20 epochs for evaluation of the simpler DL network. For both approaches the objective function was cross-entropy loss, and the models were

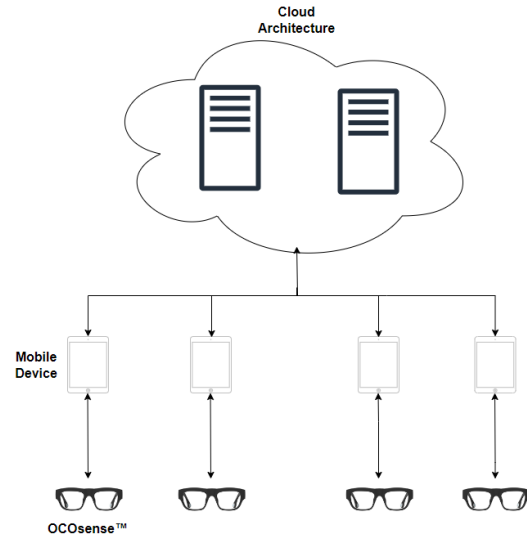


Figure 3: OCOsense™ data streaming and storage

trained using Adam optimizer. The learning rate was set to $1e-3$. For the federated learning model, the fraction fit was set to 0.7 and the fraction evaluation was set to 0.6. As an aggregating technique we employed federated averaging (FedAvg) algorithm. This evaluation aimed to shed light on any limitations or challenges that may arise when emphasizing the protection of privacy in the context of HAR.

4 RESULTS AND DISCUSSION

To evaluate the performance of each approach, we used F1-macro score. It is commonly used in multi-class classification tasks. It combines precision and recall across all classes to provide a single value that represents the classifier's average performance across the different classes. Then an unweighted mean of the F1-Scores of all classes is computed. We used this metric because it treats each class equally and prevents the dominant class from disproportionately affecting the evaluation.

4.1 Complex vs. Simple Deep Learning

To evaluate the performance of the complex and the simpler deep learning architecture 6-fold cross validation (CV) was used. This approach created participant-based folds with non-overlapping test set subjects, i.e., the data from one subject can belong to only one of the folds (either train or test) Also, it considers the unique characteristics and variations among different subjects, allowing for a more robust evaluation of the models.

The results shown in Table 1 indicate that STResNet outperformed the simpler DL model.

The complex DL model achieved an F1-Macro score of 0.87, indicating a high level of accuracy in recognizing human activities. On the other hand, the simpler DL model achieved an F1-Macro score of 0.82, demonstrating a slightly lower performance but still yielding respectable results. Interestingly, both the complex and simpler models exhibited a tendency to confuse the activities of sitting and standing. This can be attributed to the similarities in

Table 1: STResNet vs Dana

	F1-Score	Accuracy
STResNet	0.86	0.87
DANA	0.82	0.84

the IMU data collected during these two activities, which made it challenging for the models to accurately distinguish between them.

4.2 Federated vs. Centralized Learning

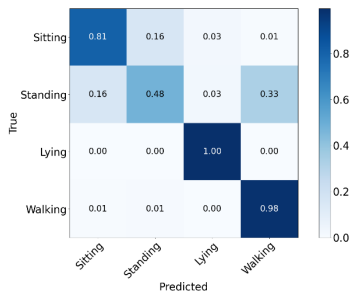
The second comparison aims to evaluate the performance of the simpler DL model alone and its federated learning implementation. The setup for both experiments was the same, for evaluating the simpler DL model we concatenated 70% of each participant's data for training and 30% of each participant's data for testing. This ensures that the results between both approaches are comparable.

The results of this comparison are shown in Table 2. The model trained using the federated learning strategy achieved a slightly lower F1-Macro score that was comparable to the simpler DL model without federated learning. This outcome suggests that the federated learning approach successfully preserved privacy while maintaining a similar level of performance to the simpler DL model.

Table 2: Centralized vs. Federated Learning

	F1-Score	Accuracy
DANA-CL	0.84	0.85
DANA-FL	0.80	0.84

From the confusion matrix (Figure 4) it can be noted that the model struggles to distinguish the sitting and standing activities. This issue emerges due to the high similarity between these two activities as observed from the perspective of a head mounted IMU. This issue extends throughout all of the experiments and warrants further investigation in future research. Lying and walking activities are predicted with a high degree of accuracy. By training the model locally on each client's data and only sharing aggregated model updates instead of raw data, the federated learning technique ensured the confidentiality and privacy of individual user data. This is particularly crucial in applications where data privacy is of paramount importance.


Figure 4: Confusion matrix of the federated learning model

5 CONCLUSION & FUTURE WORK

Overall, the results of this paper highlight the trade-off between model complexity, performance, and privacy preservation. While the complex DL model achieved slightly higher performance, it comes at the cost of increased computational complexity and potential privacy concerns. The simpler DL model, although yielding slightly lower performance, can be seamlessly integrated into a mobile app and enables real-time predictions, making it more practical for deployment in real-world scenarios. The utilization of the federated learning model in digital therapeutics offers a practical approach to leverage wearable monitoring data, enabling personalized treatments and interventions for enhanced healthcare outcomes, while preserving the privacy of the data.

In our ongoing research, we aim to explore additional techniques to further enhance the security and privacy aspects of federated learning such as advanced cryptographic protocols (homomorphic encryption, secure multi-party computation), to enable secure model updates without compromising client data privacy. Additionally, we intend to explore the potential of gossip learning, which leverages local model updates and information exchange among clients to improve scalability and robustness. Also, we plan to address the issue of misclassifying sitting and standing activities by utilizing the pressure sensor of the smart glasses. By exploring these avenues, we hope to advance the field of federated learning, making significant strides towards achieving secure and private collaborative machine learning at a scale.

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